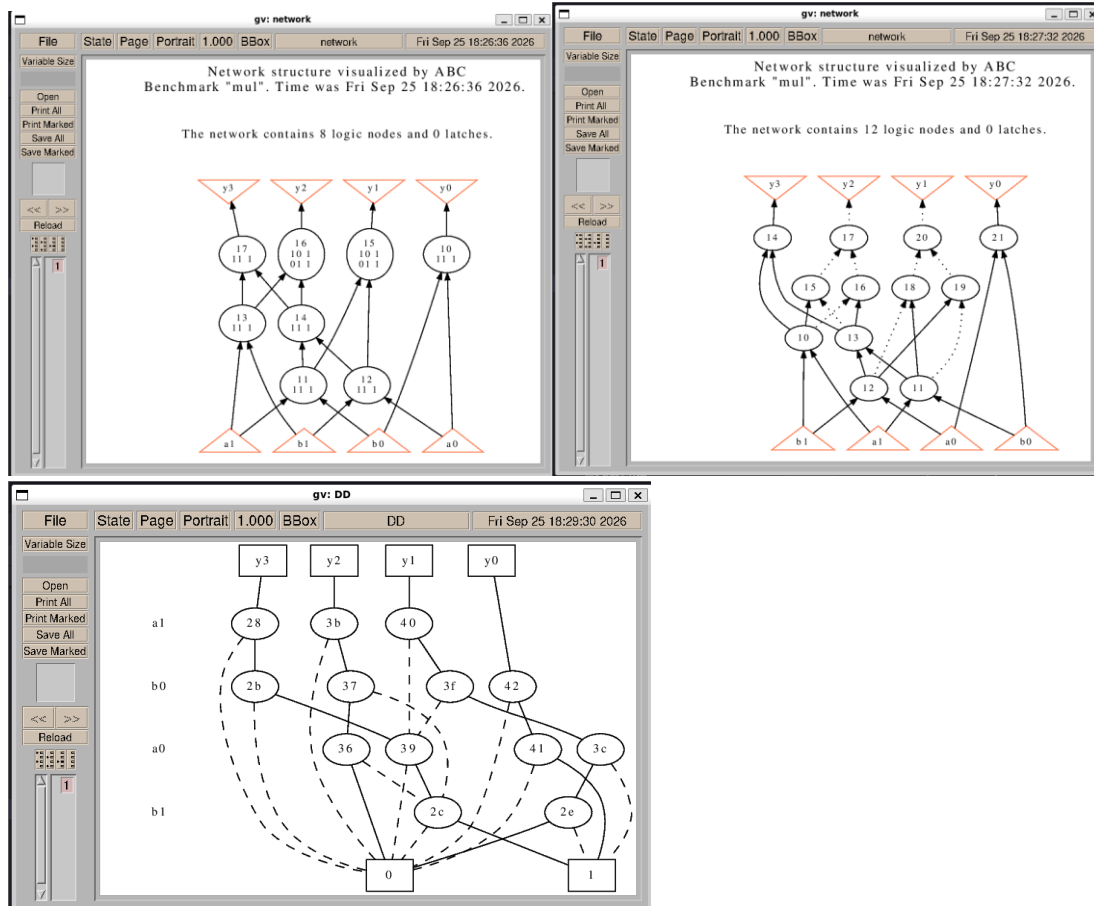


PA1-2

```
abc 01> read ./lsv/pa1/mul.blif
abc 02> print_stats
mul                               : i/o =   4/   4 lat =   0 nd =   8
edge =   16 cube =   10 lev = 3
```



PA1-3

(a)1

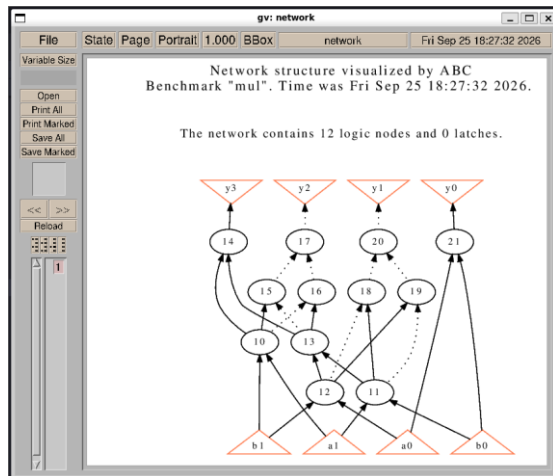
```
abc 13> aig -h
usage: aig [-h]
converts node functions to AIG
```

```
abc 13> strash -h
usage: strash [-acrih]
transforms combinational logic into an AIG
```

```
abc 13> read lsv/pa1/mul.blif
abc 14> aig
abc 14> print_stats
mul                               : i/o =   4/   4 lat =   0 nd =   8 edge =   16 aig =   12 lev = 3
abc 14> read lsv/pa1/mul.blif
abc 15> strash
abc 16> print_stats
mul                               : i/o =   4/   4 lat =   0 and =   12 lev = 4
```

For 'aig': The 8 nodes stay. Each node function is now a local AIG. aig = 12 counts the AND nodes inside those local AIGs.

For 'strash : ' The network is replaced by one structurally hashed AIG. 12 AND nodes and dotted complemented edges. The original node boundaries are gone.



(a)2

```
abc 20> bdd -h
usage: bdd [-rsh]
          converts node functions to BDD
```

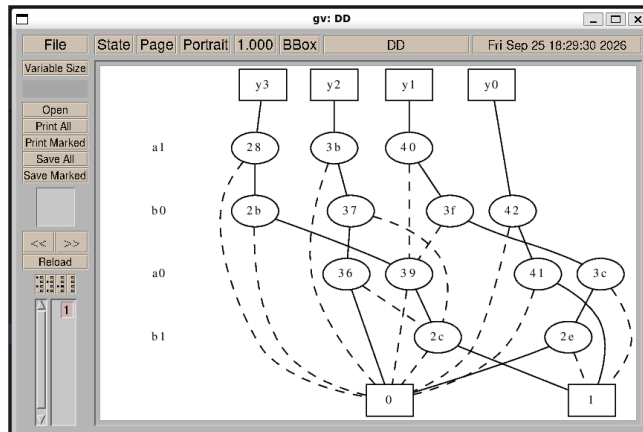
```
abc 20> collapse -h
usage: collapse [-B <num>] [-L file] [-rodxvh]
          collapses the network by constructing global BDDs
```

```
abc 17> read lsv/pal/mul.blif
abc 18> bdd
abc 18> print_stats
mul : i/o = 4/ 4 lat = 0 nd = 8 edge = 16 bdd = 16 lev = 3
```

```
abc 18> read lsv/pal/mul.blif
abc 19> collapse
abc 20> print_stats
mul : i/o = 4/ 4 lat = 0 nd = 4 edge = 14 bdd = 14 lev = 1
```

For 'bdd' : The 8 nodes stay. Each node function is now a local BDD. bdd = 16 counts the BDD nodes inside those local BDDs.

For 'collapse': The network is replaced by global BDDs, one node per output (nd = 4, lev = 1). show_bdd -g is the same for both. The output functions are the same ROBDD(canonical).



(b) Following are stats and snapshot of original blif, structurally hashed AIG, and logic network with node function in SOP. (not the same as the original blif graph)

```
abc 22> read lsv/pa1/mul.blif
abc 23> print_stats
mul          : i/o =  4/  4 lat =  0 nd =  8 edge =  16 cube =  10 lev = 3
abc 23> strash
abc 24> print_stats
mul          : i/o =  4/  4 lat =  0 and =  12 lev = 4
abc 24> logic
abc 25> print_stats
mul          : i/o =  4/  4 lat =  0 nd =  12 edge =  24 cube =  12 lev = 4
abc 25> show
```

